

Lecture 27: Bayesian MMSE

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- 1 The MMSE estimator::: {.callout-note icon=false}## Definition: Bayesian MMSEAmong *all* estimators, the one minimizing $E[(\theta - \hat{\theta})^2]$ is the posterior mean:

$$\hat{\theta}_{\text{MMSE}} = E[\theta \mid \text{data}].$$

... . ::: {.callout-warning icon=false}## Theorem: Optimality of the conditional expectation $E[\theta \mid \text{data}]$ minimizes $E[(\theta - c)^2 \mid \text{data}]$ over all constants c (given the data) – the same fact underlying the law of total variance (Lecture 10).:::—## MMSE vs. MAP vs. MLE, side by side

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—title: “Lecture 27: Bayesian MMSE”author: “Jamie Haddock”format: revealjs: output-file: Lecture27_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy.stats import beta as beta_disttheta_true, n_flips = 0.65, 15prior_a, prior_b = 2, 2flips = r
```

—# Closing the loop: our three running examples- **Randomized Kaczmarz**: discrete distributions (L3) -> convergence in probability (L15) -> Markov chains (L19).- **Randomized trace estimation**: linearity of expectation (L4) -> variance reduction (L10) -> concentration inequality (L14).- **Randomized matrix multiplication**: a derived distribution (L9) -> the optimal, variance-minimizing choice (L16) -> point estimation (L22).Each is a genuinely useful algorithm, and each turned out to be a complete tour through this course in miniature. Great final project material!That’s a wrap on new material for Math 151! See you at project presentations, and good luck on the final.## Reading & HomeworkReading: §8 (Bertsekas & Tsitsiklis, *Introduction to Probability*).